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March 2, 1973
SUPERSEDING
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FEDERAL SPECIFICATION

HOSE, FIRE, WOVEN-JACKETED RUBBER- OR CAMBRIC-LINED, WITH COUPLINGS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers woven-jacketed rubber- or cambric-lined fire hose with couplings.

1.2 Classification.

1.2.1 Types, classes, and sizes. Fire hose shall be of the following types, classes, and sizes, as specified (see 6.2):

Type I - Cotton jacket.
Type II - Cotton-polyester jacket.
Type III - Polyester jacket (class B only).

Class A - Double jacketed (types I and II only).
Class B - Single jacket.

Size 1-1/2 - 1-1/2-inch minimum inside diameter (I.D.).
Size 2 - 2-inch I.D. minimum (type III, class B only).
Size 2-1/2 - 2-9/16-inch minimum I.D.
Size 3-1/2 - 3-1/2-inch minimum I.D. (type I, class A only).
Size 4 - 4-inch minimum I.D. (type I, classes A and B, and type II, class B).
Size 4-1/2 - 4-1/2-inch minimum I.D. (type I, class A, and type III, class B only).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the specified herein:

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Federal Specifications:

WW-C-621 - Coupling Assembly, Hose, (Fire, Woven-Jacketed, Rubber- or Cambric-Lined and Unlined Linen).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 162 - Hose, Rubber, Visual Inspection Guide for
Fed. Test Method Std. No. 601 - Rubber: Sampling and Testing.
Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-775 - Hose, Rubber, or Fabric (Including Tubing) and Fittings, Nozzles and Strainers, Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-831 - Test Reports, Preparation of.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Laws and Regulations:

21CFR 121.2562 - Federal Food, Drug, and Cosmetics Act; Food Additives.

(The code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal Agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc. Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

American Society for Testing and Materials (ASTM) Standards:

D380 - Testing Rubber Hose.
D412 - Tension Testing of Vulcanized Rubber.
D2240 - Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish a fire hose for the first article inspection and approval (see 4.3, 4.3.1, and 6.4).

3.2 Labeling of hose. When specified (see 6.2), the hose shall be approved and labeled by a nationally recognized testing agency or laboratory (see 6.3) adequately equipped, and competent to perform testing of fire hose. This requirement is mandatory for Navy procurements.

3.3 Material. All material shall be new and unused. The hose shall have been manufactured not more than one year prior to date of delivery.

3.3.1 Rubber. The rubber or latex used in the lining shall be compounded from natural rubber, synthetic rubber, or a mixture of the two.

3.3.2 Jacket yarns. The yarns used in the jacket shall be cotton or polyester yarns, or a mixture thereof as required.

3.4 Construction. The hose shall consist of a lining with or without rubber backing completely covered with a woven jacket or jackets. The contour at the fold line of the finished hose shall be a smooth continuous curve and shall show no evidence of crease.

3.4.1 Hose lining. The lining for types I and II hose shall consist of a solid rubber compound or a cambric fabric coated with latex or rubber. Type III hose lining shall consist of a solid rubber compound.

3.4.1.1 Rubber lining. The rubber lining shall be uniform in thickness and shall be made from a single ply extruded tube or from calendered sheets of not less than three plies, lap-jointed and vulcanized in the round into one solid body. The thickness of rubber lining, exclusive of the rubber backing, shall be as shown in table I. The thickness shall be determined in accordance with 4.6.2. When specified (see 3.8.6, 6.1.1, and 6.2), the rubber liner shall be so compounded as to insure its conformance with the requirements of Federal Food, Drug, and Cosmetic Act, Code 21 of Federal Regulations 121.2562. The supplier shall submit a certificate attesting that the compound conforms to the above stated regulations.

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TABLE I. Thickness of rubber lining (exclusive of backing)

Size of hose	Thickness minimum (inch)
1-1/2, 2, 2-1/2	0.030
3-1/2, 4, 4-1/2	0.072

3.4.1.2 Cambric lining (types I and II). The cambric fabric shall conform to the following:

Construction	- Warp 105 ends/inch minimum, filler 85 picks/inch minimum.
Thickness of cambric fabric	- 0.007-inch minimum.
Thickness of cambric fabric with rubber	- 0.015-inch minimum.
Weight of cambric fabric	- 2.75 ounce/square yard minimum.

The inner waterway latex tube thickness shall be a minimum of 0.010-inch after removal from the cambric fabric and measured in accordance with 4.6.2.

3.4.1.3 Surface. The waterway surface of the lining shall appear free from corrugations, pitting, and other imperfections as determined by visual examination.

3.4.1.3.1 When corrugations cannot be satisfactorily determined by visual examination, hose may be subjected to the friction or head loss test of 4.6.7. If the hose meets the requirements of 3.12, the liner waterway smoothness shall be considered as meeting the requirements of this specification.

3.4.2 Rubber backing. The rubber backing, if used, shall be made from calendered sheet. The backing shall be uniform in thickness and the thickness shall be not more than 0.028-inch measured from the bottom of the impressions of the jacket.

3.4.3 Jacket. The jacket(s) shall be seamless and shall have the fillers woven around the lining throughout its length. Class A jackets may be separate or interwoven. At no point shall the liner or the backing show through the outer jacket, nor shall the filler yarn protrude through the warp yarn. The jacket(s) shall be evenly and firmly loomed and shall be free from knots, lumps, and irregularities in weave.

3.4.3.1 Type I. The jacket(s) of type I hose shall be woven from cotton yarns in both the warp and filling directions.

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3.4.3.2 Type II. The jacket(s) for type II hose shall be woven from yarns in the warp and polyester fiber yarns in the filling directions.

3.4.3.3 Type III, class B. The jacket of type III, class B hose shall be made of polyester fiber and shall be seamless and circular woven. The finished jacket for 1 1/2-inch, 2-inch, and 2 1/2-inch hose shall have a minimum of eight filler picks per inch and thirty warp ends per inch. The filler yarn shall be continuous-filament polyester fiber having a minimum twist of one turn per inch in the S direction. The warp yarn shall be spun from a suitable denier per filament and staple length of first quality polyester staple fiber, and shall have a minimum ply twist of 2 turns per inch. The jacket construction shall also meet the requirements of table II.

TABLE II. Type III jacket construction

Hose size (inch)	Warp denier per 1/ fabric cross section (minimum)	Filler yarn denier (minimum)	Count of warp 2/ single yarn (maximum)	No. of warp plys (minimum)
1-1/2	1,080,000	16,000	8	7
1-1/2	1,080,000	16,000	10	9
1-1/2	1,080,000	16,000	12	11
2	1,872,000	16,000	8	13
2	1,872,000	16,000	10	16
2	1,872,000	16,000	12	19
2-1/2	2,340,000	16,000	8	13
2-1/2	2,340,000	16,000	10	16
2-1/2	2,340,000	16,000	12	19

1/ The warp denier per fabric cross section = (No. of warp ends) (denier per warp end).

2/ The count of 840 yds/lb is a numerical value which represents the smallest acceptable yarn.

3.4.4 Inside diameter. The inside diameter of the hose shall be not less than the nominal specified size except that the 2-1/2-inch size shall have an inside diameter of at least 2-9/16 inches.

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3.4.5 Outside diameter. Unless otherwise specified (see 6.2), the outside diameter of the hose shall be at the option of the manufacturer.

3.4.6 Length. Hose shall be furnished in the length specified (see 6.2), and shall conform to table III when the hose is measured in accordance with 4.5.2.1.

TABLE III. Length of hose

Size inches	Average length per lot (min. feet)	Before sampling each length (min. feet)	After sampling Individual length (min. feet)
1-1/2, 2, 2-1/2 3-1/2, 4	50	48	47
1-1/2, 2, 2-1/2	25	24	22
4-1/2	14	13 1/2	13

3.5 Couplings and gaskets. Unless otherwise specified (see 6.2), each length of hose shall be fitted with a set, one male and one female, of couplings conforming to WW-C-621. When specified (see 6.2), both couplings shall be either male or female. Unless otherwise specified, (see 6.2), couplings shall be type A, style 1, or type C and have National Fire Hose (NH) threads. When specified (see 6.2), size 1-1/2 and 2-inch threads shall be the National Hose Coupling (NPSH) thread form with 11-1/2 threads per inch and size 4 threads shall be NH (special) thread form with 6 threads per inch. Each set of type A, style 1 couplings shall be provided with three rubber gaskets having the same temperature characteristics as the lining of the type of hose with which it is used. One gasket shall accurately fit the internally threaded swivel of the coupling, and one not less than 3/16-inch in thickness, shall be placed under the end of each expansion ring. When the coupling is made up snugly as in service, the gasket shall not be compressed to the extent that it projects beyond the inside surface of the coupling and into the waterway. Each set of type C couplings shall be provided with one gasket which shall accurately fit the swivel of the coupling. Polyvinyl chloride gaskets may be used in lieu of rubber gaskets for types I and II.

3.5.1 Type I, class A, 4-1/2 inch size. Unless otherwise specified (see 6.2), 4-1/2-inch size, type I, class A hose lengths shall be fitted with a set (one male and one female) of couplings having NH threads, and conforming to type A, style 3, of WW-C-621, except that the couplings shall be capable of withstanding the test pressures specified in table IV.

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3.6 Bursting strength. The hose shall not burst at pressures less than those shown in table IV for the respective type, class, and size of hose when tested as described in 4.6. The pressure shall be applied to the hose when it is lying straight and when it is held in a circular arc of 27-inch radius.

TABLE IV. Pressure to be applied to hose in burst test

Type	Class	Size	Burst pressure p.s.i.g.
I	A	1-1/2, 2-1/2, 3-1/2	600
II	A	1-1/2, 2-1/2	600
I and II	B	1-1/2, 2-1/2	500
I and II	B	4	300
I	A	4, 4-1/2	450 *
III	B	1-1/2, 2, 2-1/2	600
III	B	4-1/2	200

* Hose lying straight only.

3.7 Hydrostatic properties. The hose, complete with couplings, shall not leak, sweat, or show leakage at the couplings, and shall not show breakage of jacket yarns when it is subjected to the applicable hydrostatic pressures specified in table V for a minimum of 15 seconds and tested for elongation, expansion, twist, warp, and rise as described in 4.6 and 4.6.1.1. If desired, the manufacturer may humidify the hose for the hydrostatic test. Upon passing these tests, the hose shall be marked (see 3.13) to indicate compliance with the hydrostatic test requirements. The hose shall also meet table V requirements for the kink test as described in 4.6.

3.7.1 Coupling retention properties. Unless otherwise specified (see 6.2), hose assemblies shall pass the coupling retention test specified in 4.6.1.2. Any slippage of couplings shall be cause for rejection of the hose length.

3.8 Physical properties of the lining.

3.8.1 Tensile strength. The tensile strength of the lining across the fold shall be not less than 1,200 pounds per square inch (p.s.i.)(see 4.6).

3.8.2 Ultimate elongation. The ultimate elongation of the lining across the fold of types I and II hose shall be not less than 400 percent, tested in accordance with 4.6, except that the low temperature stock in type III hose shall have an elongation of not less than 300 percent.

3.8.3 Set. The set of the lining shall not exceed 25 percent after being stretched to an elongation of 300 percent for types I and II hose and 250 percent for type III hose (see 4.6).

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TABLE V. Requirements for hydrostatic tests applied to full length

	Type I							Type II					Type III		
	Class A double				Class B single			Class A double		Class B single			Class B single		
	1½	2½	3½	4&4½	1½	2½	4	1½	2½	1½	2½	4	1½	2&2½	4½
Test pressure for initial measurement, p.s.i.	10	10	10	10	10	10	10	10	10	10	10	10	10	10	10
Final hydrostatic test pressure for elongation expansion, twist, warp, rise, and coupling retention, p.s.i.g.	400	400	400	300	300	300	200	400	400	300	300	200	400	400	125
Test pressure, kink test, p.s.i.g.	350	350	250	250	300	300	200	350	350	300	300	200	350	350	100
Elongation, maximum, percent	8	8	10	14	10	10	10	8	8	10	10	10	10	10	N.A.[3]
Expansion, circumference, maximum, percent	3	3	3	3	3	3	3	6	6	6	6	6	6	6	N.A.[3]
Twist[1] to right, maximum degrees per foot	29	12½	12¼	12¼	50	25	15	29	12½	50	25	16	50	25	N.A.[3]
Twist[1] to left, maximum degrees per foot	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Warp[2], maximum inches	20	20	20	20	20	20	20	20	20	20	20	20	20	20	N.A.[3]
Rise, maximum, inches	0	0	0	0	7	4	2	0	0	7	4	2	7	4	N.A.[3]

[1]A twist to the right, the direction that would tend to tighten the couplings, is indicated by a clockwise rotation of the coupling, looking from the hose toward the coupling. A twist to the left is indicated by a counterclockwise rotation.

A maximum twist to the left of 2 degrees per foot will be permitted while pressure is being raised, but any final twist must be to the right.

[2]Deviation from a straight line.

[3]Not applicable.

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3.8.4 Strength of lap.

3.8.4.1 Rubber lining. The strength of the lap in the calendered rubber lining shall be not less than 75 pounds per inch of width (see 4.6).

3.8.4.2 Cambric latex or rubber-coated fabric. The strength of the cambric latex or rubber coated lap shall be such that the break will occur in the fabric unless the stress is greater than 75 pounds. A parting of the lap, by the shearing of the latex from the fabric without breaking the fabric, at a value under 75 pounds shall constitute a failure. The test shall be made as described in 4.6.

3.8.4.3 Hardness (type III, class B hose). When tested as described in 4.6, the hardness of the lining of type III, class B hose, after having been aged as described in 4.6.3, shall not show a Shore A Durometer reading change exceeding ± 10 points in comparison to the Shore A Durometer reading of the unaged lining.

3.8.4.4 Specific gravity (type III, class B hose). The specific gravity of material used for lining type III, class B hose shall not exceed 1.4 when tested as described in 4.6.

3.8.5 Accelerated aging. The tensile strength and elongation of the rubber lining after having been aged as described in 4.6.3 shall be not less than 70 percent of the tensile strength and elongation before aging.

3.8.6 Resistance to water (types I, II, and III hose).

3.8.6.1 Extraction of rubber lining. When specified (see 3.4.1.1, and 6.2), the extractable nonvolatile matter of the rubber liner of types I, II, and III hose shall not exceed 21 milligrams per square inch when tested as specified in 4.7.

3.8.7 Torsional stiffness and cold-tension recovery (type III, class B hose). Unless otherwise specified (see 6.2), type III, class B hose shall be subjected to the torsional stiffness and cold-tension recovery test. After low-temperature conditioning (94 hours at minus 65° F.) the torsional stiffness ratio of the rubber lining shall not exceed 10 (see 4.6.5.1 through 4.6.5.2.4) and the cold-tension recovery of the lining at minus 65° F. shall be not less than 20 percent (see 4.6.5.3 through 4.6.5.3.3).

3.9 Low temperature flexibility (type III, class B). Unless otherwise specified (see 6.2), type III, class B hose shall not show damage to the jacket or liner, and shall not leak when hydrostatically tested, after being subjected to the low temperature flexibility test (see 4.6.4).

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3.10 Resistance to fungus (types I and II). When specified (see 6.2), types I and II hose jackets, and cambric lining if used, shall be treated for resistance to fungus. The breaking strength of the cambric lining and types I and II hose jacket(s) as received shall not decrease more than 10 percent in either direction after leaching (see 4.6.6). The fungus treatment shall have no detrimental effect on the physical properties of the jacket or lining.

3.11 Adhesion. The adhesion between the backing and jacket, between the backing and lining, and between lining and jacket of types I, II, and III hose shall be not less than 12 pounds per each 1-1/2 inch of width (see table VII). This requirement is not intended to bar the construction of hose having no adhesion between the jacket and the lining along the fold provided that the surface over which there is no adhesion is not greater than 35 percent of the total surface for class A hose. Class B hose shall have the tube fully attached around the entire surface.

3.12 Friction or head loss. The first article sample shall meet the friction or head loss maximum requirements of table VI when tested in accordance with 4.6.7.

TABLE VI. Maximum allowable friction loss, p.s.i. per 50 feet

Size of Hose, inches	Flow rate Gals. per min.	Nozzle size inches	Allowable loss Per 50 ft. (p.s.i.)	Approx. Nozzle Pressure (p.s.i.)
1-1/2	100	7/8	17	19.3
2	150	1	9	26
2-1/2	250	1-1/4	8	29
3-1/2	298	1-1/2	4	20

3.13 Marking.

3.13.1 Civil agencies. Each length of hose shall be indelibly stenciled at the ends in letters at least 1 inch high with the manufacturer's name, the month and year of manufacture, the words "Tested to ___p.s.i.", and the contract number. (The applicable hydrostatic pressure specified for the size of the hose listed in table V as final pressure for elongation expansion, twist, warp, and rise p.s.i.). The lettering shall begin approximately 4 feet from the coupling. If a manufacturer produces hose at more than one factory, each length of hose shall have a distinctive marking, which may be in code, by which it may be identified as the product or a particular factory. Hose having rubber lining meeting the FDA requirements (see 3.4.1.1) shall have the following addition marking "FOR USE WITH POTABLE WATER".

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3.13.2 Military agencies. In addition to the marking required in 3.13.1, the identification marking shall be in accordance with MIL-STD-130.

3.14 Workmanship. The finished hose shall be free from jacket abrasion, knot filler, missing picks, skip warps, snags, twisted jackets, improper location of markings, or any other defect affecting appearance or serviceability. Hose shall be flexible and easily coiled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection lot. All hose lengths with couplings of the same type, class, and size with the same construction materials, produced under essentially the same conditions, and offered for inspection at one time shall be considered a lot for purposes of inspection.

4.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.3 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on one hose assembly when the first article sample is required (see 3.1). The inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be a standard production item from the supplier's current inventory provided that the unit meets the requirements of the specification and is representative of the design, construction, and manufacturing techniques applicable to the remaining units to be furnished under the contract.

4.3.1 First article test report. After completion of the first article inspection, a test report shall be prepared in accordance with MIL-STD-831, and three copies of the report shall be furnished to the procuring activity.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following:

a. Individual examination and test (see 4.4.1).
b. Sampling for tests (see 4.4.2).

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4.4.1 Individual examination and tests. Each length of hose or hose assembly shall be subjected to the following examination and tests:

(a) Visual examination and dimensional inspection, except for inside diameter (see 4.5 through 4.5.2.1).

(b) Hydrostatic pressure test (see 4.6.1.1 and 4.6.1.2).

4.4.2 Sampling for tests.

4.4.2.1 Sampling for burst test, straight and curved. A random sample of hose lengths shall be selected from each lot offered to the Government in accordance with MIL-STD-105 at inspection level S-1. From each sample selected, a 3-foot section shall be removed and tested by the applicable test methods described in table VII. Tests of the sample selections shall alternate between bursting the straight sample and the curved sample. The acceptable quality level (AQL) shall be based on 2.5 percent defective units.

4.4.2.2 Sampling for construction, accelerated aging, tensile strength, percent set, strength of lap adhesion, water resistance, hardness, specific gravity, torsion stiffness ratio tests, cold-tension recovery, and inside diameter. Samples for the jacket construction, and for accelerated aging tensile strength, ultimate elongation, percent set, strength of lap adhesion, hardness, specific gravity, torsional stiffness ratio, and cold-tension recovery of the lining may be taken from the samples used in the burst test. In the event that insufficient samples can be obtained from the burst test samples, a random sample of hose lengths shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-1. The AQL shall be 2.5 percent defective. From the hoses sampled, 3-foot sections shall be cut and tested, as specified in 4.6.2 through 4.6.5.2 (except 4.6.4).

4.4.2.3 Sampling for low temperature flexibility (type III hose only). Unless otherwise specified (see 6.2), a random sample of hose lengths shall be selected from each lot offered to the Government in accordance with tables I and II of MIL-STD-105 and inspection level S-1 for the low temperature flexibility test specified in 4.6.4. The AQL shall be 2.5 percent defective units.

4.4.2.4 Sampling for fungus resistance test. When fungus resistance is specified (see 6.2), an 8-foot section shall be taken from the manufacturer's assembled hose run in each contract for the fungus resistance requirement specified in 3.10 and as tested in 4.6.6. The test shall be repeated if any change in material occurs.

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4.5 Examination.

4.5.1 Visual examination. Each length of hose shall be examined for the defects listed in Fed. Std. No. 162. In addition to the defects listed in Fed. Std. No. 162, type III hose shall be examined to determine if the liner shows through the jacket, and if the filler yarn protrudes through the warp yarn. Any hose assembly evidencing one or more defects shall be rejected.

4.5.2 Dimensions.

4.5.2.1 Length. The distance between the inside edges of the couplings shall be determined as described in method 2411 of Fed. Test Method Std. No. 601 except that the measurement shall be made while a hydrostatic pressure of 10 p.s.i. is applied to the specimen as described in method 10311 of Fed. Test Method Std. No. 601. The lengths of all hose in the lot shall be added and averaged. The average length per lot and the individual lengths shall not be less than the corresponding lengths shown in table III.

4.5.2.2 Inside diameter. The inside diameter shall be determined as described in method 2341 of Fed. Test Method Std. No. 601 except that not less than two measurements shall be made on the unit or test unit. The measurements of hose with couplings shall be made when the 3-foot length is taken for the burst test, the 3-foot length for low temperature flexibility, or the 8-foot length for resistance to fungus.

4.6 Tests. Hose samples selected shall be subjected to the applicable tests specified in table VII.

TABLE VII. Requirement test for physical properties

Property	Requirement paragraph	Method in Fed. Test Method Std. No. 601, or ASTM	Reference paragraph or footnote
Hydrostatic pressure properties	3.7 & table V	10211	4.6.1
Kink under pressure	3.7 & table V	10361	-
Elongation under pressure	3.7 & table V	10311	-
Expansion under pressure	3.7 & table V	10321	-
Twist, degree per foot	3.7 & table V	10331	-
Warp	3.7 & table V	10341	-
Rise	3.7 & table V	10351	-
Construction	3.4	-	4.6.2 & 4.6.2.1
Bursting strength straight and curved	3.6 & table IV	10011 & 10021	- -
Accelerated aging	3.8.5	-	4.6.3

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TABLE VII. Requirement test for physical properties (contd.)

Property	Requirement paragraph	Method in Fed. Test Method Std. No. 601, or ASTM	Reference paragraph or footnote
Tensile strength of rubber liner	3.8.1	ASTM D412	1/
Ultimate elongation of rubber lining	3.8.2	ASTM D412	1/
Set of rubber lining	3.8.3	ASTM D412	1/
Strength of lap	3.8.4	4311	2/
Adhesion	3.11	ASTM D380	-
Resistance to fungus	3.10	-	4.6.6
Friction or head loss	3.12	-	4.6.7
Resistance to water extraction	3.4.1.1 & 3.8.6	-	4.7
Additional tests for type III, class B hose only			
Hardness	3.8.4.3	3021 or ASTM D2240	-
Specific gravity	3.8.4.4	14011	-
Torsional stiffness and cold-tension recovery	3.8.7	-	4.6.5
Low temperature flexibility	3.9	-	4.6.4

1/ For rubber lining, die A in ASTM D412 shall be used for cutting the specimen; for the cambric latex or rubber-coated lining, the coating on the passage way side (opposite side next to jacket) shall be carefully stripped from the fabric and die C in ASTM D412 shall be used. The specimen shall be cut transversely across the fold so that the fold is approximately midway between the ends of the constructed portion of the specimen.

2/ When testing cambric lining, the lap (splice) shall be a section that includes the fabric.

4.6.1 Hydrostatic pressure test and coupling retention.

4.6.1.1 Hydrostatic pressure test. Each length of types I, II, and III hose in the lot with coupling attached, or length of hose remaining after removal of portions on which couplings have been attached by the manufacturer for use in other tests, shall be subjected to a hydrostatic pressure test and to the tests specified in table VII. The pressure applied to the hose length shall be as specified in table V. Each length of types I, II, and III hose assemblies shall be index-marked with a pencil or other suitable device at a point immediately adjacent to the back side of each coupling. No slippage or leakage at the couplings or hose shall occur when the pressure specified in table V is applied to the length.

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4.6.1.2 Coupling retention test. Unless otherwise specified (see 6.2), one hose assembly in the lot shall be index-marked with a pencil or other suitable device at a point immediately adjacent to back side of each coupling. The hose assembly shall be subjected to half the pressure specified in table V for five minutes. The pressure shall then be raised to the normal hydrostatic test pressure for 5 seconds. No slippage or leakage shall occur at the couplings during the test. If hose assembly fails the test, the remaining hose assemblies in the lot shall be tested for 5 minutes. Failure of hose assembly to pass the test shall be cause for rejection.

4.6.2 Construction and physical property tests. The thickness of the lining and backing shall be determined as described in method 2011 of Fed. Test Method Std. No. 601. The fabric and yarns of the hose jacket (see 3.3.2 and 3.4.3 through 3.4.3.3) and the cambric (see 3.4.1.2) if used in the lining shall be taken from a 3-foot hose section (see 4.4.2.2), and shall be measured and tested in accordance with Fed. Test Method Std. No. 191 for the following characteristics:

Method	Characteristic
5041	Weight (see 3.4.1.2).
5050	Construction (see 3.4.1.2 and 3.4.3.3).
5030	Thickness (see 3.4.1.1 and 3.4.1.2).
4050 and 4052	Twist (see 3.4.3.3).
4010	Length-weight relation for determination of denier filler yarn and counts of the single yarn (see 3.4.3.3).

After removal of the specimens used in tests of the lining and backing as applicable, the remaining specimens of type III hose jacket, and the fabric used in cambric lining if used, shall be unraveled as necessary and tested for consistency.

4.6.2.1 In addition to the above test, the yarns from the sample of type III hose jacket used in test method 5050.1 shall be unraveled and counted to determine that the number of warp plies specified in 3.4.3.3 is correct.

4.6.3 Accelerated aging. The resistance of the lining to accelerated aging shall be determined as described in method 7221 or method 7231 of Fed. Test Method Std. No. 601 except that the specimens shall be exposed for 166 ± 2 hours at 70° ± 1° C. (158° ± 2° F.) for types I and II hose, and 46 ± 1/4 hours at 100° ± 2° C. (212° ± 4° F.) for type III hose. The deterioration shall be measured by change in tensile strength, elongation, and hardness. Die I in method 4111 shall be used for cutting specimens

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(see 4.4.2.2) for tensile strength and elongation, and the specimens shall be cut transversely across the fold so that the fold is approximately midway between the ends of the constructed portion of the specimen.

4.6.4 Low temperature flexibility, type III, class B. Unless otherwise specified (see 6.2), a 3-foot section of hose shall be removed from each of the hose samples selected in accordance with 4.4.2.3 and shall be exposed to a temperature of minus 65° ± 2° F. (minus 54.3° ± 1° C.) for a period of 24 hours. At the end of the exposure period and while maintained at the minus 65° ± 2° F. (minus 54.3° ± 1° C.) exposure temperature, the hose shall be rapidly bent 180° double on itself, first one way and then the other. The hose shall then be examined for evidence of cracking or breaking of the jacket and liner. The hose shall then be allowed to thaw for 24 hours at room temperature after which it shall be subjected to the applicable hydrostatic proof pressure. Leakage shall be cause for rejection.

4.6.5 Torsional stiffness ratio and cold recovery test.

4.6.5.1 Specimens. Unless otherwise specified (see 6.2), six test specimens (see 4.4.2.2) shall be buffed to uniform thickness, 0.080-inch wide by 1-1/2 inches along with a 1/4-inch square stub at each end, and tested for torsional stiffness ratio and cold tension recovery.

4.6.5.2 Torsional stiffness ratio.

4.6.5.2.1 Conditioning of specimens and test apparatus. Specimens shall be conditioned for not less than 94 hours in a chamber maintained at minus 65° ± 2° F. (minus 54.3° ± 1° C.). Test apparatus shall be conditioned for not less than 2 hours at minus 65° ± 2° F. (minus 54.3° ± 1° C.).

4.6.5.2.2 Test apparatus. The test apparatus shall be a torsional tester conforming to Fed. Test Method Std. No. 601, method 5611, except that only the torsion apparatus, the stand, and a suitable bracket to hold the lower clamp are required. Torsion wire of appropriate torsion constant shall be used.

4.6.5.2.3 Test procedures. The test procedure shall be as follows:

a. With the specimen and the test apparatus at plus 73° ± 5° F. (22.8° ± 3° C.), the lower end of the specimen shall be secured in the rigidly mounted clamp of the test apparatus and the upper end shall be secured in the upper pinch clamp attached to the lower end of the torsional wire.

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b. The pointer shall be adjusted to the zero position by rotating the protractor scale. The torsional head shall be turned 180° and the pointer reading shall be recorded after a period of 10 seconds.

c. After the above-mentioned test has been conducted, the specimen and test apparatus shall be conditioned (see 4.6.5.2.1). After conditioning is complete, the test shall be repeated on the conditioned specimen and tests shall be conducted at minus 65° ± 5° F. (54.3° ± 3° C.).

4.6.5.2.4 Calculations. The test shall be conducted on three specimens and the results averaged. If the torsional stiffness ratio for any one specimen is greater than 10, the ratio for each of the three specimens shall be within 20 percent of the average values of the three specimens. The torsional stiffness ratio at minus 65° ± 2° F. (54.3° ± 1° C.) shall be calculated as follows:

$$\text{Torsional stiffness ratio at minus 65° F.} = \frac{\text{Torsional stiffness factor at minus 65° F.}}{\text{torsional stiffness factor at 73° F.}}$$

where:

$$\text{Torsional stiffness factor} = \frac{180° - \text{angle of twist of specimen}}{\text{angle twist of specimen.}}$$

4.6.5.3 Cold-tension recovery test.

4.6.5.3.1 Test apparatus. The test instrument shall consist of a measuring board, similar to that illustrated in figure 1, on which several stretching devices can be mounted. Each stretching device shall have one movable clamp and one fixed clamp. The board shall have lines engraved at intervals corresponding to each 10 percent stretch, based on the length of the specimen between the 1/4 by 1/4-inch square stubs.

4.6.5.3.2 Test procedure. The test procedure shall be as follows:

a. The measuring baord shall be conditioned for not less than 2 hours at minus 65° ± 5° F. (minus 54.3° ± 3° C.). Three specimens, cut from the sample (see 4.4.2.3), at 73° to 78° F. (22.8° to 25.6° C.), shall be clamped in the stretching devices along the edges of stubs, adjacent to the linear part, of the specimen and laid on the board. The movable clamp shall be pulled back until the 1-1/2-inch long portion of the specimen has been stretched 100 percent elongation and fixed in that position.

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b. The stretching devices and the specimens shall be conditioned in low-temperature chamber for 7 days at minus 65° (minus 54.3° ± 3° C.).

c. With the test instrument and specimens still in the low-temperature chamber, the movable clamp shall be released from its fixed position, and the assembly shall be conditioned for an additional period of 30 minutes at minus 65° ± 5° F. (minus 54.3 ± 3° C.).

d. The final length of the specimen shall be determined with the board, stretching devices, and specimens held at an angle of 15° from the vertical. The cold-tension recovery percentages for the three specimens shall be calculated and averaged. The average value shall be used to determine compliance with 3.8.7. If the cold-tension recovery percentage for any one specimen is less than 20, the percentage for each of the three specimens shall be within 10 percent of the average for the three specimens. A cold-tension recovery not within 10 percent of the average for the three specimens shall constitute failure of the test.

4.6.5.3.3 Calculation. The precentage of cold-tension recovery shall be computed from the formula:

$$\text{Percentage cold-tension recovery} = \frac{(Ls - Lf)}{Ls - Lo} \times 100$$

where:

Ls = stretched length of specimen
Lf = final strength of specimen
Lo = initial length of specimen.

4.6.6 Fungus resistance (types I and II hoses). When specified (see 6.2), the jacket material shall be tested for resistance to fungus as described in Fed. Test Method Std. No. 191 using untreated and uncoated samples of hose jacket, and cambric lining if used, as loomed in lieu of 12- to 14-ounce cotton duck, as a reference standard. Ten samples shall be subjected to an incubation period of 4 weeks. The maximum deterioration between any sample shall not vary more than 10 percent. The procedure shall be as follows:

a. From the sample length of hose furnished, prepare two sets of ten specimens each; one set for breaking strength controls and one set for leaching, inoculation, and incubation. The lining shall be removed from the break strength control samples before cutting to size. The removal of the lining must be done in such a manner that the jacket(s) will not be incured. On cambric lined hose, the lining shall be removed in such a manner that the fabric is not injured.

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b. Cut of the specimens 1-1/2 inches wide, and 6 inches long, and then ravel the specimen to 1 inch width. The cambric lining may be cut to 1 inch width and not raveled. Control breaking strength will then be made on the jackets and on the cambric coated lining where used. The second set of specimens shall be cut not less than 1-1/2 inches wide by 8 inches long. The lining shall be left intact on these specimens during the leaching inoculation and incubation.

c. Following incubation, washing, and drying, the lining shall be carefully removed so as not to injure the jackets or the fabric in the cambric lining, where used, and to provide at least 6 inches of length of undamaged specimen.

d. After removal of the lining, the jacket specimens shall be cut 6 inches in length, and raveled to 1 inch in width. Where cambric lining is used, the specimens of lining shall be cut to 6 inches in length by 1 inch in width.

4.6.7 Friction or head loss. The friction or head loss test shall be made with the coupling gaskets in place. Two or more lengths shall be coupled together. Friction or head loss from hose inlet to outlet shall be measured with calibrated gages or a manometer or through the use of a gage and a calibrated flow meter. The pressure at the base of the standard tapered nozzle, which will produce the desired flow, shall be calculated by the following formula:

$$h = \left[1 - \left(\frac{d}{D}\right)^4\right] \times \left[\frac{Q}{19.636\ Kd^2}\right]^2$$

where

h = head at nozzle in feet of water
Q = flow in gallons per minute
d = diameter of nozzle opening in inches
D = inside diameter of hose in inches
K = discharge coefficient of nozzle (0.97 unless accurately determined).

4.6.7.1 The maximum allowable head loss shall be no greater than the values shown in table VI.

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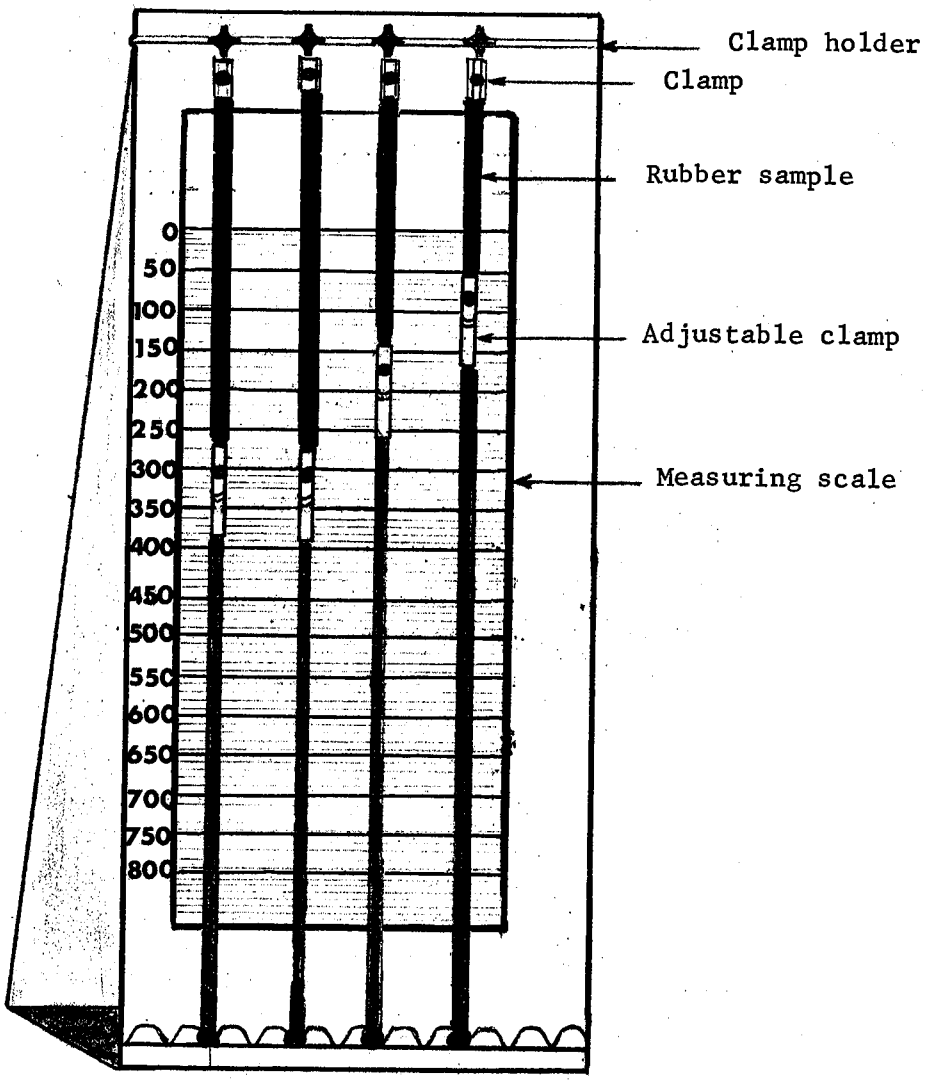


Figure 1. Tension recovery apparatus.

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4.7 Extraction of inner hose liner. When specified (see 3.4.1.1, 3.8.6, and 6.2) the inner tube liner surface of one hose sample, selected in accordance with 4.4.2.2, shall be extracted in distilled water at reflux temperature in accordance with the following procedure:

a. Remove a 1-inch ring of rubber inner liner and strip, cut, or buff away the outer fabric from the latex or rubber coat until the surface is smooth.

b. Cut sufficient length of inner rubber liner to obtain approximately 10 grams.

c. Subject sample to distilled water, at reflux temperature, for 7 hours in an extraction apparatus as shown in method 16001 of Fed. Test Method Std. No. 601.

d. Filter the solution through #40 filter paper and collect the extract in a tared container.

e. Extract the rubber sample in an additional 50-75 cubic centimeters of distilled water at a reflux temperature for 2 more hours and repeat step d. Combine the extract solutions and evaporate to dryness. The container and residue shall be dried in an oven at 105° ± 3° C. (221° ± 5° F.) for one hour, cooled in a desiccator and weighed. The weight of the residue shall be recorded. A blank shall be run using the same amount of distilled water. After making allowance for the blank, the weight of the residue shall be recorded to the nearest milligram

f. Report the amount of matter extracted from the rubber in milligrams per square inch of surface, determined as follows:

Length (inches) x width (inches) x 2 - total surface area (sq.in.)

Milligrams per square inch = weight of extract/total surface area.

4.8 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the hose shall be inspected to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The preservation and packaging shall be level A or C, as specified (see 6.2).

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5.1.1 Level A. The hose shall be preserved and packaged in accordance with level A of MIL-H-775. When hose is subjected to hydrostatic tests at the manufacturer's plant, the hose shall be carefully drained and the jackets thoroughly dried before preparation for shipment.

5.1.2 Level C. The hose shall be cleaned, preserved and packaged in a manner that will afford protection against corrosion, deterioration, mechanical and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his own practice providing it fulfills these requirements. When hose is subjected to hydrostatic tests at the manufacturer's plant, the hose shall be carefully drained and the jackets thoroughly dried before preparation for shipment.

5.2 Packing. The packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The hose shall be packed in accordance with level A of MIL-H-775.

5.2.2 Level B. The hose shall be packed in accordance with level B of MIL-H-775.

5.2.3 Level C. The hose shall be packed in a manner that will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Class A hose is intended for use on pumping engines and in places where service conditions require the additional protection against wear by the extra jacket. Types I and II, class B hose is intended for use at fire hydrants, stand-pipes, reels, and places where abrasion of the jacket is not considered severe. Type III, class B hose is intended for general use and for use on fire fighting equipment.

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6.1.1 Special use. Hoses covered by this specification may be used for water purification equipment and for the conveying of potable water when the Federal Food, Drug Cosmetic Act requirements have been met.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Type, class, size, and length required (see 1.2.1 and 3.4.6).

(c) If first article is not required (see 3.1, 4.3 and 6.4).

(d) If testing by an independent laboratory is required (see 3.2 and 6.3).

(e) If inner rubber liner is required to conform to Federal Food, Drug, and Cosmetic Act (see 3.4.1, 3.8.6.1, 4.7 and 6.1.1).

(f) If a particular outside diameter of hose is required (see 3.4.5).

(g) If couplings are other than as specified and whether both couplings shall be male or female (see 3.5).

(h) When thread couplings shall be NPSH (see 3.5 and 3.5.1).

(i) If no or different coupling retention properties are required (see 3.7.1 and 4.6.1.2).

(j) If torsional stiffness test is not required for type III hose (see 3.8.7 and 4.6.5.1).

(k) If low temperature flexibility is not required for type III hose (see 3.9, 4.4.2.3, and 4.6.4).

(l) If fungus-resistance treatment is required (see 3.10, 4.4.2.4, and 4.6.6).

(m) If coupling retention test is not required (see 4.6.1.2).

(n) Location and conditions for first article testing (see 4.1, 4.3, and 6.4).

(o) Level of preservation and packaging, and level of packing required (see 5.1 and 5.2).

(p) Marking required (see 5.3).

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6.2.1 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9 (n) of the Armed Services Procurement Regulation, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9 (n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DD 1664
4.3.1	Report, First Article Test	UDI-T-24009

(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 Recognized testing agency or laboratory. The testing agency or laboratory should be one which is regularly engaged in the examination, testing, and evaluation of fire hose, which has an established factory inspection, listing, and labeling program; and which has published standards for listing and labeling that are nationally recognized. For example, the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611 and the Factory Mutual Engineering Division, 1151 Boston-Providence Turnpike, Norwood, MA 02062 are considered recognized testing agencies.

6.3.1 It should be noted that all applicable requirements of this specification should be verified in accordance with section 4. However, hose bearing the label of a recognized testing agency or laboratory will require additional testing only for those requirements, as applicable, which have not been performed or confirmed by the agency label service.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be the first production item and shall consist of one hose assembly. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test, and approval of the first article.

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Navy - YD
Air Force - 84

Review activity:

Navy - SH

User activities:

Army - MI
Navy - MC, CG

Preparing activity:

Navy - YD

CIVIL AGENCY COORDINATING ACTIVITIES:

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